WESTMINSTER
International University in Tashkent

System Analysis & Design

Tutorial 02

Agenda

WIUT

1. The role of Diagrams in SAD
2. Data Flow Diagram (DFD)
3. Activity Diagram (AD)

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Diagrams

The role of diagrams in SAD.

The role of Diagrams

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- Assume we need to document requirements for 2 different cases:
  1. Classes in an information system.
  2. Describe what should be done when button X is clicked.

- What method of documenting would you choose: Text or Diagram?
- Why?

- Can you document anything with:
  A. Diagram
  B. Text

The role of Diagrams

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- An image/diagram worths a thousand words.
- Diagrams simplify the complex information and makes it easier to understand.
- Stakeholders may not understand tech documentation but can quickly grasp a diagram.
- Reduce misunderstanding of the requirements (DFD, ERD, etc.).
- Help detecting redundancies, inconsistencies or missing information.
- Create a single communication language and increase communication efficiency.
- Makes communication across teams, organizations or time possible.

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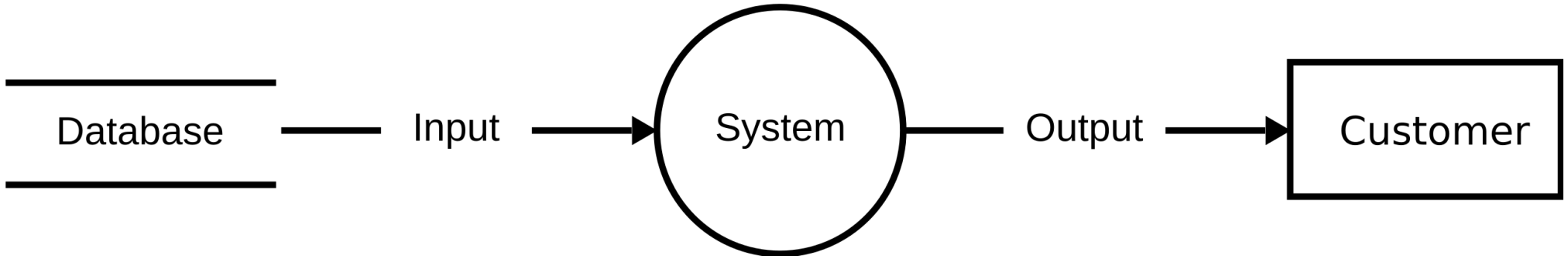
Data Flow Diagram

Documenting how data flows through a system.

Data Flow Diagram

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Example:



Source: https://en.wikipedia.org/wiki/Data-flow_diagram

What is Data Flow Diagram?

Google

"A Data Flow Diagram (DFD) is a visual map that illustrates how data moves through an information system or business process, using standardized symbols to show data inputs, outputs, transformations, and storage points."

Source: https://www.google.com/search?q=data+flow+diagram

Wikipedia

"A DFD is a way of representing a flow of data through a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow — there are no decision rules and no loops."

Source: https://en.wikipedia.org/wiki/Data-flow_diagram

What is Data Flow Diagram?

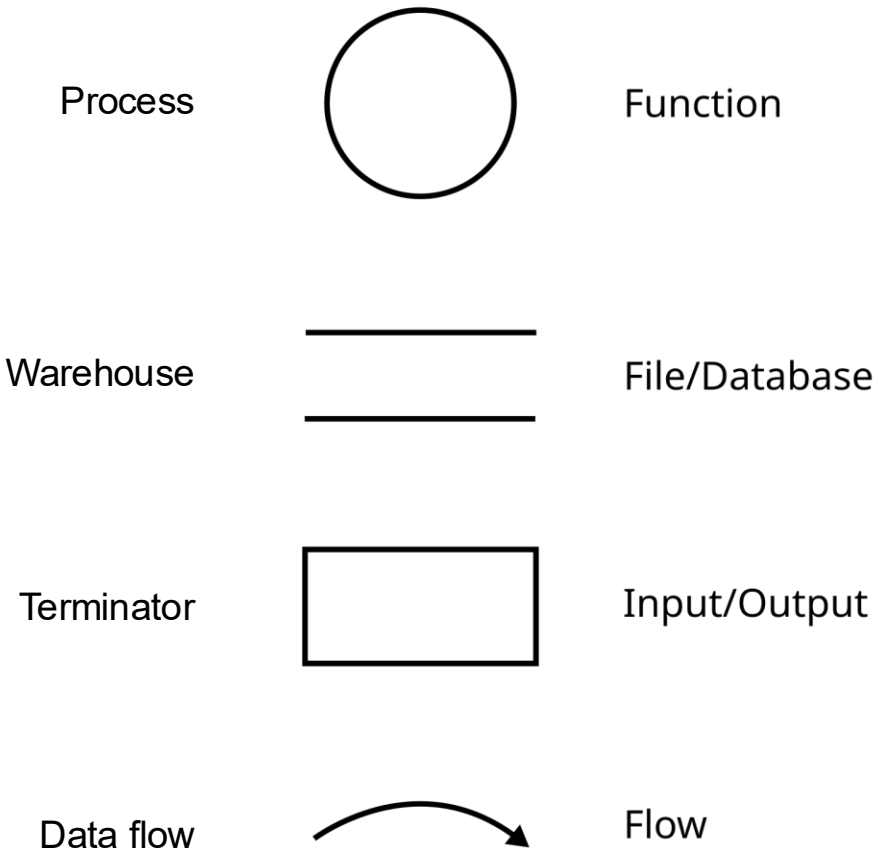
A Data Flow Diagram is a visual representation of how data moves through a system.

It shows:

- **Terminator** – external entity that communicates with the system (people, other systems).
- **Warehouse** – warehouse used to store the data for later (file, database).
- **Process** – a process transforms inputs into outputs (system functions).
- **Data flow** – shows how data flow between from one part of the system to another.

DFD does not show any logic or implementation details — only the flow of information!

Data Flow Diagram components



Levels of DFD

- **Level 0** – High level view of the entire system as a single diagram. Documents the whole system showing all external inputs/outputs and the main system processes.
- **Level 1** – Explains one or more process from Level 0 in more detail.
- **Level 2, 3, etc.** – Dive deeper into the processes.

Exercise: Group

1. Download “SAD-Tutorial-02-Cases.pdf” file from the Intranet.
2. Read the first case.
3. Draw the DFD diagram for it on the whiteboard.

Exercise: Individual

1. Take 15 minutes to read and draw the DFD diagram for the second case (on your own).

2. Anyone would like to draw their version on the whiteboard?

3. Discuss how your diagram is similar and different from the one on the whiteboard.

Note: You can draw it on paper or use any online tool.

DFD: Final notes

1. DFD does not deal with the timing of data flows.

2. It also does not deal with events.
   For example, it does not document when any data flow is triggered.

3. Timing and events are documented using other diagrams (sequence diagram, flowchart, etc.).

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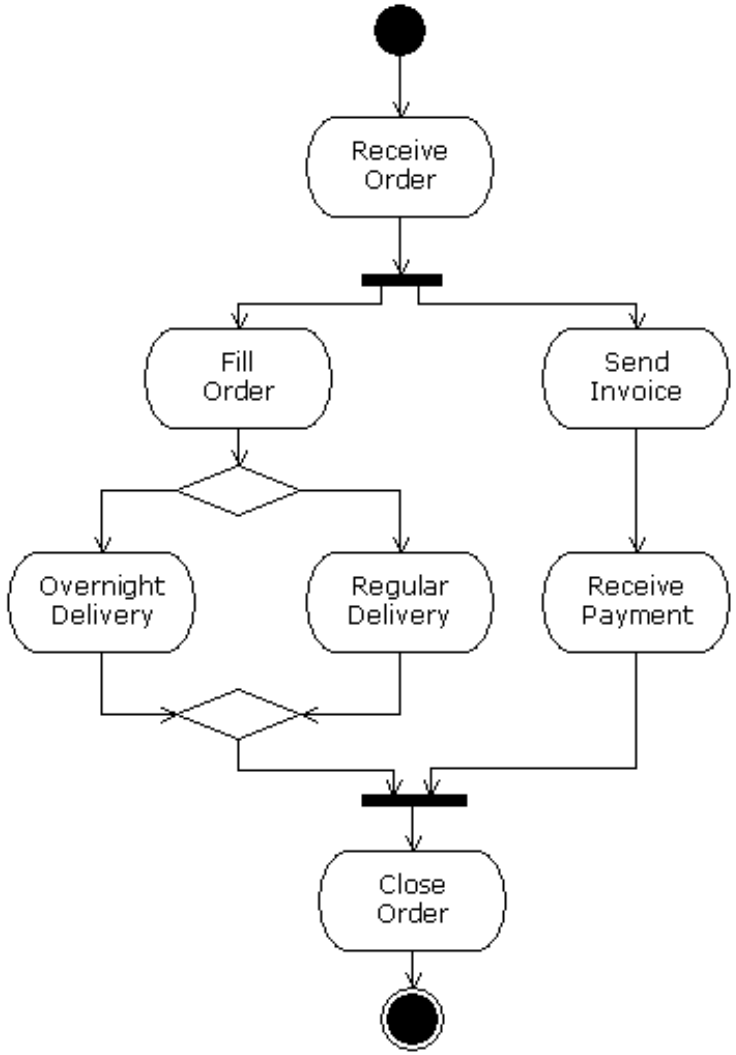
Activity Diagram

Documenting system activities and flow of control.

Activity Diagram

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Example:



Source: https://bjohnson.lmu.build/

What is Data Flow Diagram?

Google

"An activity diagram is a Unified Modeling Language (UML) diagram that visually represents the flow of control and data from one activity to the next in a system or process."

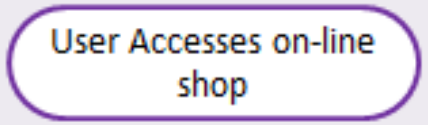
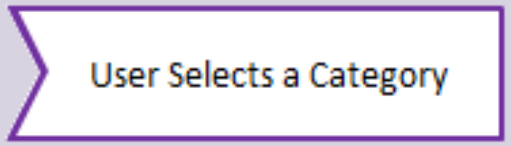
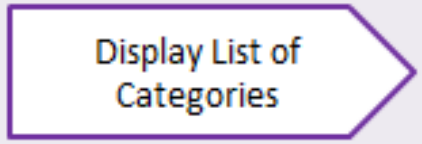
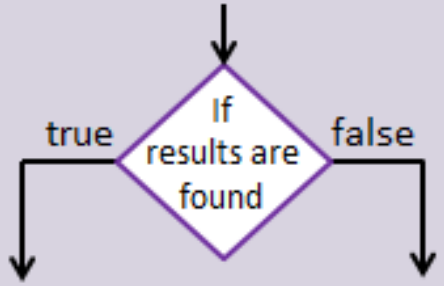
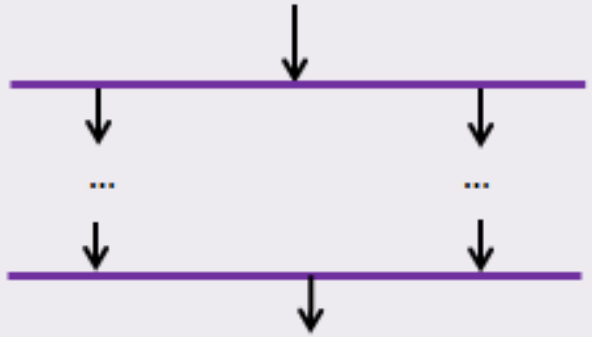
Source: https://www.google.com/search?q=activity+diagram

Wikipedia

"Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration, and concurrency. They are intended to model both computational and organizational processes (i.e., workflows), as well as the data flows intersecting with the related activities."

Source: https://en.wikipedia.org/wiki/Activity_diagram

Activity Diagram: Shapes

Activity Diagram / Shape:	Meaning:
	Starting Point
	Activity
	Input
	Output
	Decision
	Simultaneous Activities
	End Point

Exercise: Group

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Exercise: Individual

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The End

See you at the lecture.